

Sharp reversal spectra for affine prime-pair factorization

Jonas Whidden

Research draft, 28 September 2026

Abstract

We identify the exact double-logarithmic upper growth rate of maximal separated descent–ascent counts for reciprocal-smooth prime-factor laws. Every fixed rank has an attained finite maximum. We prove the arithmetic application to products of distinct affine forms in the sum of two primes, including all finite local exceptions. The rate is a spectrum of actual bounded-prime-gap frequencies and has a finite phase profile. We also prove an exact threshold equivalence between recurring short gaps and late ascents in the two canonical laws. The maintained Lean proofs use only standard foundational axioms. This AI-assisted research draft records statements and proof structure; independent review and novelty assessment remain separate work.

The source and theorem ledger are available in the public repository. The first-difference and symmetric-polynomial method builds on Wang–Crapis [1]; quantitative prime clusters provide an established analytic input [2]. The spectrum and affine-product application are separate developments whose novelty remains subject to review.

1 Objects and theorem

Index the primes by p_i , starting at $i = 0$. For fixed local probabilities η_p , define

$$D_k(p) = \eta_p[z^{k-1}] \prod_{q < p} (1 - \eta_q + \eta_q z).$$

We consider the ordinary law $\eta_p = 1/p$ and the generic odd law $\eta_2 = 0$, $\eta_p = (p-2)/(p-1)^2$ for odd primes. The defining input-size density limit precedes the limit in rank k .

A witness consists of $d < a$ with $D_k(p_{d+1}) < D_k(p_d)$ and $D_k(p_a) < D_k(p_{a+1})$. Witnesses are strictly separated: $a_i + 1 < d_j$ for $i < j$. Let $N(k)$ be the attained maximum number.

Theorem 1. *For both canonical laws, there is a common $a > 0$ such that for every $\epsilon > 0$ and all sufficiently large k ,*

$$\exp(\exp(ak)) \leq N(k) \leq \exp(\exp((1/3 + \epsilon)k)).$$

The implementation takes $a = 1/602$.

Each fixed-rank sequence eventually strictly decreases. This theorem concerns growing ranks, not infinitely many reversals at one fixed rank.

2 Proof structure

Put $w_q = \eta_q/(1 - \eta_q)$ and let E_r be the elementary symmetric polynomial in the weights below p . With supported degree and positive denominators, the sign of the step from p to the next prime p' is the sign of

$$\frac{E_{r-1}}{E_r} - \left(\frac{1}{\eta_{p'}} - \frac{1}{\eta_p} + 1 \right).$$

The threshold is $p' - p + 1$ ordinarily, and for generic odd primes is

$$p' - p + 1 - \frac{p' - p}{(p - 2)(p' - 2)}.$$

Exceptional small primes are handled before using these formulas.

Writing $S = \sum_{q < p} w_q$ and W_{r-1} for the largest $r - 1$ weights, the bounds

$$\frac{r}{S} \leq \frac{E_{r-1}}{E_r} \leq \frac{r}{S - W_{r-1}}$$

combine with $S = \log \log p + O(1)$ and a uniform bound for W_{r-1} . Above $\log \log p = (1/3 + \epsilon)k$, they force strict descent. An eventual nonincreasing tail bounds every separated witness family; a finite maximum is attained.

For the lower bound, the quantitative prime-window input supplies at least $X^{1/3}$ starts in $[X, 2X]$ whose width-600 windows contain two primes, for all sufficiently large X . The source extraction and its external analytic artifacts are pinned and audited. Bare infinitude of bounded prime gaps is insufficient here.

Factorial cells provide a preceding long gap. A bounded-fiber argument and ordered thinning give genuine consecutive-prime indices with the required strict disjointness. The uniform sign criteria turn each selected pair of long and short gaps into a reversal. More generally, an eventual X^δ supply of fixed width H , with $\delta > 0$, gives the lower bound for every $0 < a < 1/(H + 1)$. All loss constants are fixed before k grows; they are absorbed on the double-exponential scale. Taking $H = 600$, $\delta = 1/3$ and $a = 1/602$ closes the displayed theorem. The chosen width is a conservative formally closed input, not a literature claim about the smallest available gap.

3 Late ascents and recurring prime gaps

A late ascent at coefficient $b > 0$ means consecutive primes $p < q$ and a rank k satisfying

$$\log \log(p - 1) \geq bk, \quad D_k(q) > D_k(p).$$

Unbounded supply means that such a pair exists for every prescribed lower bound on *both* k and p .

Theorem 2. *Let $H \geq 1$ be an integer and $1/(H + 2) < b < 1/(H + 1)$. Infinitely many consecutive prime gaps are at most H if and only if for every K, Q there exist $k \geq K$ and consecutive primes $p < q$ with $p \geq Q$ and $\log \log(p - 1) \geq bk$, at which both the ordinary and generic odd rank densities strictly ascend.*

More precisely, suppose gaps at most $h \geq 1$ recur arbitrarily far out, and every sufficiently late gap is at least h . Then $1/(h + 1)$ is the critical coefficient: every smaller positive coefficient has unbounded simultaneous ascent supply; above every larger coefficient, both densities strictly descend for all sufficiently large ranks and lower primes. No assertion at the boundary coefficient is included.

Here is the mechanism behind the converse. For any fixed integer $H \geq 0$ and $b > 1/(H + 1)$, choose

$$\frac{1}{H + 1} < a < b.$$

The uniform symmetric-ratio estimates give $E_{k-2}/E_{k-1} \leq 1/a$ at sufficiently large k, p with $\log \log(p - 1) \geq bk$. Since $1/a < H + 1$, the ordinary threshold $q - p + 1$ forbids an ascent whenever $q - p \geq H$. In the odd law its correction is bounded by $1/(p - 2)$, which is absorbed by the same fixed strict margin.

Thus unbounded ascent supply forces recurring gaps strictly below H . Apply this with $H + 1$ to obtain gaps at most H in the equivalence.

For the forward direction choose a scale $a > b$ with $(H + 1)a < 1$, and assign ranks to arbitrarily late short gaps so that $\log \log(p - 1)/k$ tends to a . The lower symmetric-ratio bound tends to $1/a > H + 1$, forcing both ascents. The rank-rounding and finite starting exceptions disappear inside the fixed strict margins. For the upper critical statement, use the eventual lower bound $q - p \geq h$; for its lower statement, use recurrence at most h .

In particular, for every fixed $b > 1/5$, a sufficiently late ascent in either canonical law must cross a twin-prime gap. Indeed, the converse with $H = 4$ excludes every gap at least four, and odd-prime gaps are positive even integers. An independently proved unbounded supply would therefore imply infinitely many twin primes. This conditional implication is proved; its supply hypothesis is open. It yields no unconditional new gap bound, and one finite example cannot certify the supply.

4 Exact spectrum and every fixed rank

Let $G_H(X)$ count consecutive prime gaps at most H whose lower prime is at most X , and put

$$\gamma_H = \limsup_{X \rightarrow \infty} \frac{\log \log(3 + G_H(X))}{\log \log X}, \quad \Lambda(\nu) = \sup_{H \geq 2} \frac{\gamma_H}{H + \nu}.$$

The cutoffs X may be taken over the natural numbers.

Theorem 3. *Fix $\nu > 0$ and a prime probability law $0 \leq \eta_p \leq 1$ such that*

$$\eta_p^{-1} = p/\nu + b + o(1)$$

along the primes. Every fixed-rank density sequence is eventually nonincreasing and has an attained finite maximal reversal count $N_\eta(k)$. Moreover,

$$\limsup_{k \rightarrow \infty} \frac{\log \log(3 + N_\eta(k))}{k} = \Lambda(\nu) > 0.$$

All parameters of the law are fixed before k grows. This is a limsup statement; the weaker assumption $\eta_p = \nu/p + O(p^{-2})$ alone is not its hypothesis.

Finite forced coordinates give exact rank shifts, and excluded coordinates can be erased. On prime prefixes, the interior odds sum is $\nu \log \log p + O(1)$

and the largest r odds have total $O(1 + \log r)$. These estimates yield uniform growing-rank ratios even at arbitrarily large prime cutoffs.

For the lower spectrum, choose a rate below $\gamma_H/(H + \nu)$. Infinitely late actual gap counts supply short gaps in a suitable double-exponential band. Preceding factorial-cell descents, bounded fibers and ordered thinning retain the rate and give strictly separated witnesses. The deleted initial segment and fixed multiplicity losses are kept before taking limits. For the upper spectrum, retain each short gap's individual cutoff. The finite envelope has the form

$$N_\eta(k) \leq T_k + \sum_{2 \leq g \leq J} G_g(U_g(k)), \quad \log \log U_g(k) \leq \left(\frac{1}{g + \nu} + \epsilon \right) k.$$

The initial term and the fixed finite sum are absorbed before letting ϵ tend to zero. This yields the matching upper rate.

For a fixed degree $r \geq 1$, write $M_r(p)$ for its Bernoulli prefix mass. The same finite bounds, with the exact forced-coordinate cases, give $M_{r-1}(p) \leq M_r(p)/2$ eventually. At consecutive sufficiently large primes $p < q$, the exact signed identity is

$$D_{r+1}(q) - D_{r+1}(p) = \eta_p \eta_q [M_{r-1}(p) - (\eta_q^{-1} - \eta_p^{-1} + 1)M_r(p)].$$

Reciprocal control makes the threshold at least $1/2$ eventually. The degree-zero recurrence handles rank one. Hence all fixed ranks, including eventual zero tails, are eventually nonincreasing.

The quantitative window input gives a class with $\gamma_H = 1$. Let H_* be the least such class. The spectrum is attained among $2 \leq H \leq H_*$ for every fixed $\nu > 0$. Since there are only finitely many smaller classes and each has exponent less than one, for all sufficiently large fixed ν ,

$$\Lambda(\nu) = \frac{1}{\nu + H_*}, \quad \frac{1}{\Lambda(\nu)} - \nu = H_*.$$

No numerical value of H_* is asserted. The condition $\gamma_H = 1$ is a double-logarithmic limsup condition; it does not assert positive density, nor a fixed positive-power lower bound at every large cutoff.

5 Actual affine prime-pair densities

Fix a nonempty family

$$F(p, q) = \prod_{j=1}^m (c_j(p + q) + d_j),$$

where c_j are positive integers, d_j are integers, and $c_i d_j \neq c_j d_i$ for $i \neq j$. Sample all ordered prime pairs $p, q \leq X$, including the diagonal, with denominator $\pi(X)^2$.

Theorem 4. *For every fixed positive rank k and prime ℓ , the density of ℓ as the k th smallest distinct prime factor of $|F(p, q)|$ exists. Its sequence over ℓ has finite maximal reversal counts at every rank, and its exact upper growth rate is $\Lambda(m)$.*

For each fixed modulus, normalized prime input counts become uniform over the units, while nonunit input classes vanish. Exact unit-pair CRT counting therefore proves joint divisibility independence for every fixed finite set of primes. Summing its atoms by cardinality gives the finite Bernoulli rank formula. Zero outputs satisfy $p+q \leq \sum_j |d_j|$ and contribute a vanishing finite correction.

At every prime, the local probability is the exact count of unit pairs annihilating F , divided by $(\ell - 1)^2$. Outside the finite coefficient, shift and determinant exceptions,

$$\eta_\ell = \frac{m(\ell - 2) + z}{(\ell - 1)^2}, \quad \eta_\ell^{-1} = \frac{\ell}{m} - \frac{z}{m^2} + o(1),$$

where $z = 1$ if one shift is zero, and $z = 0$ otherwise. Thus the general theorem applies with $\nu = m$ without discarding any exceptional prime. The input-size limit is taken first at each fixed family, rank and prime; the rank limsup follows separately for that fixed family. This concerns representation-weighted outputs, not uniform integers.

6 Formal statement and remaining research

The maintained general and arithmetic assemblies prove the reciprocal-smooth spectrum and the affine prime-pair application. Challenge states the canonical bound and arithmetic spectrum directly over Mathlib primitives; Solution contains their real proofs. The theorem ledger below identifies their exact declaration names and source owners. The local audit compares both compiled types in separate environments and checks twelve theorem closures, including the late-ascent equivalence and the conditional inverse twin-prime transfer. The deliberate Challenge placeholders do not enter the Solution proof closures.

The pinned Erdos 690 dependency supplies narrow ordinary-density leaves; its conditional classification is not promoted. The dependency is

never rebuilt or redistributed here: its published cache is reused directly. Shared Windows Lean 4.34.0 artifacts are used by probes and maintained builds. Additional analytic ports are hash-pinned, but their upstream compatibility publication remains pending; the Erdos cache alone does not provide a complete fresh-machine setup. Official Comparator/NanoDa verification is separate from the local compiled-type check.

The repository provides Challenge, Solution, the affine assembly, the recurrence equivalence, a theorem and provenance ledger, and build instructions. These branch links are mutable navigation links. The verification record identifies the source and artifact hashes used for the initial publication. This work used AI assistance for research, proof development and exposition; no independent expert mathematical review is claimed.

Independent late-ascent supply beyond inherited prime-gap inputs, the numerical spectrum, complete generic odd unimodality classification, and minimal regularity needed for finite fixed-rank reversal counts remain research questions. Exact ownership cancellation is a separate direction. No new prime-gap bound, twin-prime theorem, moving-tail Dickman law, Goldbach representation lower bound or RH energy contraction is claimed. The spectrum and arithmetic application are potentially new results; negative literature searches and successful formal builds do not establish publication significance.

References

- [1] S. Wang and D. Crapis, *A Complete Answer to Erdos Problem 690*, arXiv:2605.08542v1 (2026).
- [2] J. Maynard, *Dense clusters of primes in subsets*, Compositio Mathematica 152 (2016), doi:10.1112/S0010437X16007296; arXiv:1405.2593v2.

Copyright 2026 Jonas Whidden. Original text: Apache License 2.0 or CC BY 4.0, at the recipient's option.